

Efthymia (Efi) Tsamoura, PhD

Current Role: Technical Expert, Huawei Labs, UK.

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🌐 <https://tsamoura.github.io>

Employment and work achievements

2025 – now

📌 Technical Expert in Huawei Labs, UK.

- ☐ I have developed *commercial* techniques based on probabilistic logical reasoning for on-device data cleaning and integration for Huawei smartphones.
- ☐ I developed on-device agents based on probabilistic logical reasoning and neurosymbolic learning for email importance prediction for Huawei smartphones (*under commercial deployment*).
- ☐ I have developed new semantics for probabilistic logic programming, called *equivalence semantics*, to support the integration of logic programs with neural embedding models without falling into local minima during learning [NeurIPS 2025], improving the baseline performance by more than 42%.
- ☐ I have co-authored a survey article summarizing the latest theoretical advances on neurosymbolic symbol grounding [JAIR 2026].
- ☐ I have served as a reviewer for grant application proposals on neurosymbolic AI and LLMs from the [Leverhulme Trust](#).
- ☐ I have delivered **four keynote talks** in top-tier workshops and conferences on symbolic reasoning and neurosymbolic learning [NILA@IJCAI 2026], [ICWE 2026], [TGD@EDBT 2026], [TPM@UAI 2025].
- ☐ I have served as the Industrial Track Chair of VLDB 2026.

2019 – 2025

📌 Senior Researcher in Samsung AI Center (SAIC) Cambridge, UK.

- ☐ I led research on reasoning and neurosymbolic learning within SAIC-Cambridge.
- ☐ I developed provably correct techniques that scaled logical reasoning and query answering to billions of facts [AAAI 2020], [VLDB 2021], [SIGMOD 2023] and led to deep models with substantially higher accuracy than the state-of-the-art [ICML 2023], [AAAI 2023].
- ☐ I was the first to prove that we can learn neural classifiers using logical theories as the source of supervision [NeurIPS 2023], addressing a long-standing problem in weakly supervised and neurosymbolic learning. I was also the first to characterize learning imbalances in neurosymbolic learning [NeurIPS 2025].
- ☐ I developed CPU-based techniques capable of scaling rule mining to millions of facts in less than 1% of the runtime needed by the state-of-the-art GPU-based approaches [AAAI 2023], [arXiv 2025].
- ☐ I was the first to show how to efficiently answer queries in second-order logical theories that include millions of facts, while preserving the correctness and completeness of the answers [AAAI 2018], [arXiv 2026].

Employment and work achievements (continued)

2016 – 2018  **Research Fellow, Turing Institute, UK.**

- ☐ I was the principal investigator of a project on accessing distributed NHS data under formal privacy guarantees, receiving funding of £100K from the Turing and the FARR institutes.
- ☐ I developed provably correct techniques for accessing remote data under formal privacy guarantees [KR 2020], [TODS 2021].
- ☐ I developed the first technique for efficiently answering queries over Web-scale knowledge graphs under complex logical theories [AAAI 2018] under guarantees on the correctness and completeness of the answers.

2013 – 2016  **Postdoctoral Researcher, University of Oxford, UK.**

- ☐ I theoretically established whether and how to answer queries subject to access restrictions and data constraints under guarantees on the completeness of the answers [PODS 2014], [TODS 2016], [Morgan & Claypool 2016].
- ☐ I led a project on accessing data from the European Bioinformatics Institute (EBI), the National Centre for Biotechnology Information (NCBI), and the Kyoto Encyclopedia of Genes and Genomes (KEGG), being awarded an EPSRC IAA grant.

Older posts

Jan 2012 – May 2013  **Project manager & Research assistant**, Data Engineering Laboratory, Department of Computer Science, Aristotle University of Thessaloniki, Greece.

- ☐ I led a nationally funded project on querying data hosted in cloud infrastructure.

Nov 2008 – May 2013  **PhD studies**, Data Engineering Laboratory, Department of Computer Science, Aristotle University of Thessaloniki, Greece.

- ☐ I worked on answering queries in distributed settings under optimality guarantees and multi-objective optimization.

Oct 2007 – Jun 2008  **Research assistant**, Multimedia Knowledge Laboratory, Information Technologies Institute, Centre of Research and Technology Hellas (CERTH), Greece.

- ☐ I developed techniques for video understanding.

Jun 2006 – Sep 2007  **Research assistant**, Artificial Intelligence and Information Analysis Lab, Department of Computer Science, Aristotle University of Thessaloniki, Greece.

- ☐ I developed computer vision techniques for assembling artifacts.

Education and awards

Education

2013  **PhD**, Aristotle University of Thessaloniki, Greece.
Thesis title: *Query Optimization in Distributed and Dynamic Environments*.

2007  **BSc** in Computer Science, Aristotle University of Thessaloniki, Greece.
Grade: 9.15/10 (*highest grade in the class*).

Awards and distinctions

- 2024  **Recognized as a future leader in AI** by the Royal Society, London, being invited to the Frontiers of Science (FoS) meeting that took place in Ottawa, Canada.
- 2017  **Awarded funding of £100K** by the Turing Institute, UK, and the FARR Institute, UK, to lead a project on privacy-preserving access to data from the National Health Service.
- 2016  **Early Career Research Fellowship**, the Alan Turing Institute, UK (*selected among more than 300 applicants*).
- 2010  **Best PhD student in Computer Science**, Aristotle University of Thessaloniki, Greece.
- 2007  **Highest ranked BSc student in Computer Science**, Greek State Scholarships Foundation.
- 2006  **Highest ranked BSc student in Computer Science**, Greek State Scholarships Foundation.
 **Young scientists scholarship**, Onassis Foundation.
- 2005  **Highest ranked BSc student in Computer Science**, Greek State Scholarships Foundation.
- 2004  **Highest ranked BSc student in Computer Science**, Greek State Scholarships Foundation.

Funding

Obtained research grants

- Jan 2017  **FASHIN: Federated Access to Secure Healthcare INformation.**
Budget: *£100K.*
Role: *Principal Investigator.*
Funding source: *Turing Institute, UK, & FARR Institute, UK.*
Project aims: *Develop techniques for accessing distributed data from the National Health Service (NHS) in an efficient and privacy-preserving fashion.*
Grant applicants: Grant application authored by me.
- Dec 2015  **Simple Unified Access to Biological Resources on The Web.**
Role: *Research associate.*
Funding source: *EPSRC Impact Acceleration.*
Project aims: *Develop a platform for accessing data from the European Bioinformatics Institute (EBI), the National Centre for Biotechnology Information (NCBI), and the Kyoto Encyclopaedia of Genes and Genomes (KEGG) in a transparent fashion.*
Grant applicants: Grant application co-authored by me and Professor Michael Benedikt from the University of Oxford.

Mentoring and teaching activities

Mentoring and supervision

- 2026  **Haoran Bao**, Huawei Labs, UK (intern).
- 2025 – 2026  **Aaditya Naik**, University of Pennsylvania, USA.
Jointly with Professor Mayur Naik.
- 2024 – 2025  **Jaron Maene**, KU Leuven, Belgium.
Jointly with Professor Luc De Raedt.

Mentoring and supervision (continued)

- 2023 – 2024  **Liwei Che**, Rutgers University, USA.
Jointly with Professor Vladimir Pavlovic.
- 2022 – 2025  **Dominic Phillips**, University of Edinburgh, UK.
- 2022 – 2024  **Kaifu Wang**, University of Pennsylvania, USA.
Jointly with Professor Dan Roth.
- 2021 – 2024  **Jonathan Feldstein**, University of Edinburgh, UK.
Jointly with Dr. Vaishak Belle.
- 2021 – 2022  **Modestas Jurčius**, University of Edinburgh, UK.
- 2021  **Davide Buffelli**, Samsung AI, Cambridge, UK (intern).
- 2017 – 2018  **Benjamin Spencer**, Turing Institute, UK (postdoctoral researcher).
- 2017  **Ugo Comignani**, University Claude Bernard Lyon 1, France.
Jointly with Professor Angela Bonifati.

External PhD examiner

- Jan 2024  **Emile van Krieken**, Vrije Universiteit Amsterdam, the Netherlands.
Thesis: *Optimisation in Neurosymbolic Learning Systems*.

Teaching

- Oct 2015 – Dec 2015  *Databases (undergraduate level)*, Department of Computer Science, University of Oxford.
Role: *Tutor & Lab demonstrator*.
- Oct 2012 – Jan 2013  *Databases (undergraduate level)*, Department of Computer Science, Aristotle University of Thessaloniki.
Role: *Teaching assistant*.
- Feb 2012 – Jun 2012  *Graph Theory (undergraduate level)*, Department of Computer Science, Aristotle University of Thessaloniki.
Role: *Teaching assistant*.
-  *Design of Algorithms (undergraduate level)*, Department of Computer Science, Aristotle University of Thessaloniki.
Role: *Teaching assistant*.
- Oct 2011 – Jan 2012  *Databases (undergraduate level)*, Department of Computer Science, Aristotle University of Thessaloniki.
Role: *Teaching assistant*.
-  *Analysis of Algorithms (undergraduate level)*, Department of Computer Science, Aristotle University of Thessaloniki.
Role: *Teaching assistant*.
- Oct 2010 – Jan 2011  *Databases (undergraduate level)*, Department of Computer Science, Aristotle University of Thessaloniki.
Role: *Teaching assistant*.
- Sep 2010 – Jan 2011  *Analysis of Algorithms (undergraduate level)*, Department of Computer Science, Aristotle University of Thessaloniki.
Role: *Teaching assistant*.

Other scientific activities

Invited talks & panels

- Aug 2026  **(Keynote) Eight Years of Weakly Supervised Neurosymbolic Learning: Main Results and Algorithms**, 1st Workshop on Neuro-Symbolic Intelligence for LLMs and Autonomous Agents, co-located with the 35th International Joint Conference on Artificial Intelligence (IJCAI-ECAI 2026).
<https://sites.google.com/view/nila-workshop-ijcai-2026/home>.
- Jun 2026  **(Keynote) Scalable Rule Mining Under Formal Guarantees**, 26th International Conference of Web Engineering (ICWE 2026).
<https://icwe2026.webengineering.org/index.php/keynotes/>.
- Mar 2026  **(Keynote) Trigger Graphs & Probabilistic Equivalence: Towards Scalable and Efficient Neurosymbolic Learning and Inference**, 2nd Workshop on Transforming Graph Data, co-located with 29th International Conference on Extending Database Technology (EDBT 2026).
<https://transforming-graph-data.github.io/2026/>.
- Feb 2026  **Imbalances in Neurosymbolic Learning: Characterization and Mitigation**, UCL Centre for AI.
<https://www.eventbrite.co.uk/e/ucl-ai-centre-seminar-series-dr-efthymia-tsamoura-huawei-labs-tickets-1978960764920>.
- Jul 2025  **(Keynote) Imbalances in Neurosymbolic Learning: Characterization and Mitigation**, 8th Workshop on Tractable Probabilistic Modeling, co-located with the 41st Conference on Uncertainty in Artificial Intelligence (UAI 2025).
<https://tractable-probabilistic-modeling.github.io/tpm2025/>.
- Feb 2025  **Training Neural Classifiers Using Logic: Results, Challenges, and Applications**, Department of Computer Science, University of Oxford.
<https://www.cs.ox.ac.uk/seminars/2718.html>.
-  **Training Neural Classifiers Using Logic: Results, Challenges, and Applications**, École Nationale Supérieure de l'Électronique et de ses Applications.
- Aug 2024  **(Keynote) Training Neural Classifiers Using Logic**, 1st International Workshop on Logical Foundations of Neuro-Symbolic AI, co-located with the 33rd International Joint Conference on Artificial Intelligence (IJCAI 2024).
<https://sites.google.com/view/lnsai2024/>.
- Apr 2024  **Materializing Knowledge Bases via Trigger Graphs**, KU Leuven, Leuven, Belgium.
- Mar 2024  **On Training Neural Classifiers Using Logic: Learnability, Error Bounds and Applications**, 1st Workshop on the Foundational Aspects of Neurosymbolic Computing (FANeSy), Santiago, Chile.
<https://sites.google.com/view/fanesy-2024/main>.
- Feb 2024  **(Royal Society) Addressing the Limitations of Deep Learning**, UK-Canada Frontiers of Science on Artificial Intelligence, Royal Society of Canada, Ottawa, Canada.
<https://rsc-src.ca/en/events/frontiers-science-2024>.
- Jan 2024  **On Learning Latent Models with Multi-Instance Weak Supervision**, Vrije Universiteit Amsterdam, the Netherlands.
<https://blogs.ed.ac.uk/he-lab/2023/11/16/%e3%80%90bayes-coffee-house-tech-talk-series%e3%80%91on-learning-latent-models-with-multi-instance-weak-supervision/>.
- Nov 2023  **On Learning Latent Models with Multi-Instance Weak Supervision**, University of Edinburgh, UK.
<https://blogs.ed.ac.uk/he-lab/2023/11/16/%e3%80%90bayes-coffee-house-tech-talk-series%e3%80%91on-learning-latent-models-with-multi-instance-weak-supervision/>.

Invited talks & panels (continued)

- Oct 2023  **On Learning Latent Models with Multi-Instance Weak Supervision**, University of Cambridge, UK.
<http://talks.cam.ac.uk/talk/index/206044>.
- Jun 2023  **Highly Scalable Reasoning over Knowledge Bases via Trigger Graphs**, 2nd Annual Symposium of the Turing Interest Group on Knowledge Graphs, London, UK.
<https://github.com/turing-knowledge-graphs/meet-ups/blob/main/symposium-2023.md>.
-  **(Keynote) Reasoning at Scale: Why, How and What's Next**, 20th Extended Semantic Web Conference.
<https://2023.eswc-conferences.org/keynote-speakers/>.
- Dec 2022  **Highly Efficient Reasoning via Trigger Graphs**, 6th Stream Reasoning Workshop, Amsterdam, the Netherlands.
<https://streamreasoning.org/events/srw2021/>.
- Nov 2021  **Samsung AI Forum Panel Discussion**.
<https://research.samsung.com/2021saif>.
- Feb 2021  **Neurosymbolic Integration: A Compositional Perspective**, 1st Samsung AI Neurosymbolic Workshop.
<https://research.samsung.com/sanw2021>.
-  **Knowledge Graphs**, Samsung AI boot camp.
- Nov 2017  **Goal-Driven Query Answering for Existential Rules with Equality**, Semmler, Oxford, UK.
- Oct 2017  **Magic and other Tricks for Goal-Driven Query Answering**, University Claude Bernard Lyon 1, Lyon, France.
- Jun 2016  **Integration, Optimization, and Reasoning over Web Services**, Columbia University, New York, USA.
-  **Integrating Biological Data: Challenges and Solutions**, University of Cambridge, UK.
- Oct 2015  **Integrating Biological Data: New Opportunities and Road Ahead**, European Bioinformatics Institute, Cambridge, UK.
- Apr 2013  **Optimizing Queries in Modern Infrastructure: Challenges, Solutions and Open Problems**, Imperial College London, UK.

Guest lectures and seminars

- Aug 2025  **Neurosymbolic Learning: An Introductory Course to Theory and Applications**, 36th European Summer School in Logic, Language and Information.
<https://2025.esslli.eu/courses-workshops-accepted/course-information.html>.
- Nov 2023  **High Performance Reasoning over Knowledge Bases via Trigger Graphs**, Knowledge Graphs, MSc program, University of Edinburgh.
- Feb 2023  **Probabilistic reasoning at Scale: Trigger Graphs to the Rescue**, Machine Reasoning, MSc program, University of Pennsylvania.
- Sep 2021  **Logic & Privacy**, Samsung AI, Cambridge, UK.
- Nov 2020  **On Neurosymbolic Integration**, Samsung AI, Cambridge, UK.
- Nov 2019  **Beyond the Grounding Bottleneck: Datalog Techniques for Inference in Probabilistic Logic Programs**, Samsung AI, Cambridge, UK.
- Jan 2019  **Reasoning 2.0**, Samsung AI, Cambridge, UK.

Workshop & panel organization

- Apr 2026  **Unifying Concept Representation Learning**, 14th International Conference on Learning Representations.
<https://ucrl-iclr26.github.io>.
- Sep 2025  **1st Workshop on New Ideas for Large-Scale Neurosymbolic Learning Systems**, 51st International Conference on Very Large Data Bases.
<https://karmaresearch.github.io/NILS2025/>.
- Aug 2024  **4th Knowledge-Infused Learning Workshop**, 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining.
<https://kil-workshop.github.io>.
- Nov 2023  **Neurosymbolic AI Panel**, 22nd International Semantic Web Conference.
<https://iswc2023.semanticweb.org/neuro-symbolic-ai-panel/>.
- Aug 2023  **3rd Knowledge-Infused Learning Workshop**, 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining.
<https://aiisc.ai/kiml2023/>.
- May 2023  **Neurosymbolic AI for Agent and Multi-Agent Systems Workshop**, 22nd International Conference on Autonomous Agents and Multiagent Systems.
<https://albertmeronyo.github.io/nesymas23/>.
- 2023  **Interest Group in Neurosymbolic AI**, Turing Institute, UK.
<https://www.turing.ac.uk/research/interest-groups/neuro-symbolic-ai>.
- Jun 2022  **2nd Samsung AI Neurosymbolic Workshop**.
<https://research.samsung.com/sanw>.
- Nov 2021  **Knowledge Representation for Hybrid & Compositional AI Workshop**, 18th International Conference on Principles of Knowledge Representation and Reasoning.
<https://krhcai.github.io/>.
- Feb 2021  **1st Samsung AI Neurosymbolic Workshop**.
<https://research.samsung.com/sanw2021>.

Editorial & reviewing activities

- 2025  **Guest editor** of the special issue Neurosymbolic AI: First-Order Logic and Beyond, Journal of Neurosymbolic Artificial Intelligence.
-  **Top Reviewer**, 39th Annual Conference on Neural Information Processing Systems (NeurIPS 2025).

Software

- 2015  PDQ [VLDB 2015] [GitHub]: Java engine to support querying distributed data sources under different access patterns (e.g., web services) and database constraints (e.g., inclusion dependencies). The engine has been demonstrated in multiple scenarios, including [querying biological data available via web services](#). The foundations of PDQ have been presented in PODS 2014 and TODS 2016. [THEMES: [DISTRIBUTED DATABASES](#)]
- 2017  CHASEBENCH [PODS 2017] [GitHub]: Tools and datasets for benchmarking different techniques for querying and materializing knowledge graphs. [THEMES: [KNOWLEDGE GRAPHS](#), [REASONING](#)]

Software (continued)

- 2018  CHASEGOAL [AAAI 2018] [Webpage]: Technique to support sound and complete query answering over knowledge graphs under Datalog rules with existential quantification and equalities. CHASEGOAL outperforms by several orders of magnitude in terms of runtime and memory query answering under the commercial engine R^DFox, making the difference between answering a query in a few seconds or not being able to process the query at all. [THEMES: [KNOWLEDGE GRAPHS](#), [REASONING](#), [QUESTION ANSWERING](#)]
- 2020  vPROBLOG [AAAI 2020] [BitBucket]: First efficient C++ engine to support sound and complete query answering over probabilistic knowledge graphs using Datalog rules under the [distribution semantics](#). vPROBLOG was the first engine to outperform the state-of-the-art engine ProbLog2 by several orders of magnitude in terms of runtime, making the difference between answering a query in a few seconds or not being able to process the query at all. vPROBLOG relies on the notion of [provenance semirings](#) to enable probabilistic reasoning and database optimization techniques to speed up reasoning. [THEMES: [KNOWLEDGE GRAPHS](#), [PROBABILISTIC REASONING](#), [QUESTION ANSWERING](#)]
- 2021  NEUROLOG [AAAI 2021] [BitBucket]: Prototype for neurosymbolic integration implemented in PyTorch. NEUROLOG outperformed all state-of-the-art neurosymbolic engines available at the time in accuracy and runtime, namely DeepProbLog, ABL, and NeurASP, supporting training on datasets to which those techniques could not scale. NEUROLOG laid the foundations for formalizing the problems of [training neural classifiers using supervision from logical theories](#) and [imbalanced neurosymbolic learning](#). [THEMES: [NEUROSYMBOLIC LEARNING](#), [MACHINE LEARNING](#)]
-  GLOG [VLDB 2021] [GitHub]: C++ engine to support sound and complete reasoning over knowledge graphs using Datalog rules with existentials. GLOG outperforms commercial and open-source engines, such as VLog, R^DFox, Vadalog, WebPIE, and Inferray, by several orders of magnitude in terms of runtime, while incurring equal or lower main-memory overhead. Regarding R^DFox, GLOG substantially outperforms it in terms of time and memory efficiency despite the fact that GLOG reasons using a *single thread only*: in on-device reasoning, GLOG is more than *18x faster* (1s vs 18.7s) than R^DFox when R^DFox uses 1 thread, and up to *6x faster* (1s vs 6s) when R^DFox uses 16 threads. Furthermore, the memory requirements of GLOG are usually lower than those of R^DFox. In terms of scalability, GLOG allows the materialization of knowledge graphs with 17B facts in less than 40 min using a single machine with commodity hardware. Other knowledge graphs in which GLOG outperforms the state-of-the-art are ChaseBench, LUBM, UOBM, DBpedia, Claros, Reactome, and Yago. GLOG can be deployed on mobile phones under the Android NDK for reasoning running exclusively on the device. [THEMES: [KNOWLEDGE GRAPHS](#), [REASONING](#), [ONDEVICE](#)]
- 2022–2023  LTGs [SIGMOD 2023] [GitHub]: C++ engine to support sound and complete query answering over probabilistic knowledge graphs using Datalog rules under the [distribution semantics](#). LTGs is an extension of GLOG. LTGs substantially outperforms engines, such as ProbLog2, vProbLog, and Scallop, both in terms of runtime and memory overhead, even without approximations. The technology behind LTGs extends the notion of [provenance circuits](#). LTGs can be deployed on mobile phones under the Android NDK for reasoning running exclusively on-device. LTGs offers a [Python API](#). [Here](#) are further instructions and scripts to reproduce the experiments in the [SIGMOD paper](#). LTGs can be applied to on-device visual question answering and outperforms LXMERT and RVC on the [VQAR benchmark](#). In addition, its [Python API](#) offers easy integration with foundation models. [THEMES: [KNOWLEDGE GRAPHS](#), [PROBABILISTIC REASONING](#), [COMPUTER VISION](#), [VISUAL QUESTION ANSWERING](#), [MACHINE LEARNING](#), [ONDEVICE](#)]

Software (continued)

- NGP [AAAI 2023] [Access upon request]: NGP is a technique implemented in PyTorch for regularizing deep models for scene graph generation (SGG) at training-time, by injecting commonsense knowledge. NGP achieves the following: (i) it improves the accuracy of IMP, MOTIFS, and VCTree baseline SGG models, by up to 33%, leading to 16% absolute accuracy improvements when applied in conjunction with TDE; (ii) it outperforms GLAT and LENSr, two state-of-the-art training-time regularization techniques, by up to 18% and 15%; (iii) it can improve the accuracy of a baseline SGG model by up to six times when restricting the availability of ground-truth training facts; (iv) it outperforms in accuracy BGNN by up to 90% and KBFN by up to 86% when combined with TDE. NGP has been evaluated on the Visual Genome and the Open Images v6 datasets. This library builds upon Mask R-CNN for bounding box detection. [THEMES: COMPUTER VISION, SCENE GRAPH GENERATION, MACHINE LEARNING, NEUROSymbOLIC LEARNING]
- 2023 ■ CONCORDIA [ICML 2023] [GitHub]: Teacher-Student neurosymbolic framework, where instead of being a complex deep model, the teacher is a probabilistic logical theory. The framework has been implemented in PyTorch. CONCORDIA supports supervised, semisupervised and unsupervised training and has been applied to a variety of tasks, exceeding the relevant state-of-the-art. In particular, CONCORDIA outperforms DPL, Bi-LSTM, and DistilBERT on entity linking and IARG and PSL-CAD on collective activity detection when using MobileNet and Inception-v3 as backbone networks. CONCORDIA is strictly more expressive than DPL and T-S in terms of the types of supported logical theories. [THEMES: COMPUTER VISION, NEUROSymbOLIC LEARNING, VIDEO ACTIVITY DETECTION, MACHINE LEARNING, LANGUAGE MODELS]
- PRISM [AAAI 2023] [GitHub]: Mining rules from relational data is a key problem in AI. The basis of rule mining is the mining of repeated data patterns, known as *structural motifs*. Despite the importance of extracting “good” structural motifs, the problem was not well understood. PRISM is the first principled technique for mining structural motifs for learning languages that blend first-order logic with probabilistic models. PRISM is implemented in C++ and can improve over state-of-the-art rule mining techniques, namely LSM and BOOSTr, by up to 6% in terms of accuracy and up to 80% in terms of runtime. PRISM has been the basis for developing SPECTRUM, an even more efficient C++ rule mining technique that scales to datasets containing millions of facts. Furthermore, SPECTRUM outperforms state-of-the-art rule mining techniques for entity linking, such as AMIE3, RNN-Logic, and NCRL, on the most well-known benchmarks (such as WN18RR and FB15K-237) by up to a factor of 100 in terms of runtime, while running exclusively on CPU. [THEMES: RULE MINING, KNOWLEDGE GRAPHS, MACHINE LEARNING, LINK PREDICTION]
- 2024 ■ SO-QA [arXiv 2026] [Webpage]: Supporting sound and complete query answering over knowledge graphs under very complex classes of rules has been a long-standing problem in AI. This is the first technique to support query answering under second-order rules with equality on the order of milliseconds. The technique has been implemented on top of RD-Fox’s Java API and has been developed in collaboration with Oxford Semantics Technologies. [THEMES: KNOWLEDGE GRAPHS, REASONING, QUESTION ANSWERING]
- 2025 ■ EQUALITY-REASONING [NeurIPS 2025] [GitHub]: The integration of logic programs with embedding models resulted in a class of neurosymbolic frameworks that jointly learn symbolic rules and representations for the symbols in the logic. To support this integration without falling into local optima during learning, we revamped the well-known possible world semantics of probabilistic logic programs into new semantics called equivalence semantics. Experiments on well-known benchmarks show that the equivalence semantics leads to neurosymbolic models with up to 42% higher performance than the Neural Theorem Prover (NTP), the Greedy Neural Theorem Prover (GNTP), the Conditional Theorem Prover (CTP), and DeepSoftLog. [THEMES: NEUROSymbOLIC LEARNING, EMBEDDING MODELS]

Software (continued)

- CAROT [[NeurIPS 2025](#)] [[GitHub](#)]: Mitigating learning imbalances (a problem typically referred to as long-tailed learning) has received considerable attention in supervised and weakly supervised learning with the proposed techniques operating at training or at test time. However, these previous algorithms are not appropriate for neurosymbolic learning. We propose a statistically consistent technique for estimating the marginals of the hidden labels given weak labels and two algorithms to mitigate imbalances during training and test time. The first algorithm assigns pseudolabels to training data based on a novel linear programming formulation of neurosymbolic learning. The second algorithm uses the marginals of the hidden labels to constrain the model's predictions on test data using robust semi-constrained optimal transport. Our techniques improve accuracy over strong baselines in neurosymbolic–[Semantic Loss](#) and [Scallop](#)– and long-tailed learning– [Logic Adjustment](#) and [RECORDS](#)– by up to 14%.
[THEMES: [NEUROSYMBOLIC LEARNING](#)]

Research publications

Symbols ($\alpha\beta$) and (*) are used to denote alphabetical ordering of the authors and joint first authorship, respectively.

Books

- ($\alpha\beta$) Michael Benedikt, Julien Leblay, Balder ten Cate, and **Efthymia Tsamoura**. “*Generating Plans from Proofs: the Interpolation-Based Approach to Query Reformulation*”. Synthesis Lectures on Data Management. Morgan & Claypool Publishers, 2016. isbn: 978-3-031-00728-6. [[Link](#)]

arXiv Preprints

- Aaditya Naik*, **Efthymia Tsamoura***, Shibo Jin, Mayur Naik, and Dan Roth. “*On Improving Neurosymbolic Learning by Exploiting the Representation Space*”. arXiv:2602.07973. 2026. [[arXiv](#)]
- **Efthymia Tsamoura** and Boris Motik. “*Goal-Driven Query Answering over First- and Second-Order Dependencies with Equality*”. arXiv:2412.09125. 2026. [[arXiv](#)]
- Jonathan Feldstein, Dominic Phillips, and **Efthymia Tsamoura**. “*Efficiently Learning Probabilistic Logical Models by Cheaply Ranking Mined Rules*”. arXiv:2409.16238. 2025. [[arXiv](#)]
- Jonathan Feldstein, Paulius Dilkas, Vaishak Belle, and **Efthymia Tsamoura**. “*Mapping the Neuro-Symbolic AI Landscape by Architectures: A Handbook on Augmenting Deep Learning Through Symbolic Reasoning*”. arXiv:2410.22077. 2024. [[arXiv](#)]

Conference Proceedings

- Stefania Dumbrava and **Efthymia Tsamoura**. “*Rule-Based Languages for Neurosymbolic AI*”. In: Proceedings of the 10th International Joint Conference on Rules and Reasoning (RuleML+RR), 2026 (*to appear*).
- Jaron Maene and **Efthymia Tsamoura**. “*Embeddings as Probabilistic Equivalence in Logic Programs*”. In: Proceedings of the 39th Conference on Neural Information Processing Systems (NeurIPS), 2025. [[pdf](#)]
- **Efthymia Tsamoura***, Kaifu Wang*, and Dan Roth. “*Imbalances in Neurosymbolic Learning: Characterization and Mitigating Strategies*”. In: Proceedings of the 39th Conference on Neural Information Processing Systems (NeurIPS), 2025. [[pdf](#)] [[arXiv](#)]
- Kaifu Wang, **Efthymia Tsamoura**, and Dan Roth. “*On Learning Latent Models with Multi-Instance Weak Supervision*”. In: Proceedings of the 37th Conference on Neural Information Processing Systems (NeurIPS), 2023. [[pdf](#)] [[arXiv](#)]

- Jonathan Feldstein, Modestas Jurcius, and **Efthymia Tsamoura**. “*Parallel Neurosymbolic Integration with Concordia*”. In: Proceedings of the International Conference on Machine Learning (ICML), Honolulu, Hawaii, USA, 23-29 July. pp. 9870–9885. 2023. [[pdf](#)] [[arXiv](#)]
- **Efthymia Tsamoura**, Jaehun Lee, and Jacopo Urbani. “*Probabilistic Reasoning at Scale: Trigger Graphs to the Rescue*”. In: Proceedings of the ACM SIGMOD 2023 International Conference on Management of Data (SIGMOD), Seattle, WA, USA, 12-23 July. 2023. [[pdf](#)] [[arXiv](#)]
- Jonathan Feldstein, Dominic Phillips, and **Efthymia Tsamoura**. “*Principled and Efficient Motif Finding for Structure Learning of Lifted Graphical Models*”. In: Proceedings of the 37th AAAI Conference on Artificial Intelligence (AAAI), Washington, DC, USA, February 7-14, pp. 12205–12215. 2023. [[pdf](#)] [[arXiv](#)]
- Davide Buffelli* and **Efthymia Tsamoura***. “*Scalable Theory-Driven Regularization of Scene Graph Generation Models*”. In: Proceedings of the 37th AAAI Conference on Artificial Intelligence (AAAI), Washington, DC, USA, February 7-14. pp. 6850–6859, 2023. [[arXiv](#)]
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